

MATH 61, UCLA, Fall 2025

Introduction to Discrete Structures

Instructor:

Daniel Soskin

- Location:
 - Bunche Hall 1209B
- Time:
 - Monday, Wednesday, Friday, 11am-11:50am
- Office: MS 6236
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- Office Hours:
 - Wednesday 5pm-6pm

Teaching Assistants:

John Zhou

- Dis 1A
 - Kaufman Hall 101
 - Tues 11am-11:50am
- Dis 1B
 - Kaufman Hall 101
 - Thur 11am-11:50am
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- Office Hours: Friday 14:00-15:00

William Chang

- Dis 1C
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Norman Chen

- Dis 1E
 - Bunche Hall 3153
 - Tues 11am-11:50am
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- Office: MS 2903
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Text. R. Johnsonbaugh, *Discrete Mathematics*, (8th edition), Prentice-Hall.

Course Information. Lecture, three hours; discussion, one hour. Discrete structures commonly used in computer science and mathematics, including sets and relations, permutations and combinations, graphs and trees, induction.

Prerequisites. Courses 31A, 31B. Not open for credit to students with credit for course 180 or 184.

Course structure.

- There will be 5 Quizzes during your Discussion sessions, roughly every other week. The lowest quiz score will be dropped, but there will be no make up quizzes given.
- You can take quizzes only in the Discussion section in which you are registered. *Quizzes taken in other Discussion sections will not be graded.*
- There are 2 Midterm exams and one cumulative Final exam. These will be held in person. If you must miss a midterm for any reason, you will automatically default to Grading Scheme 2.
- The midterms will be held in class on Friday, October 24 and Wednesday, November 19.
- The Final exam will be held on Friday, December 12 from 3pm-6pm, location TBA.
- You will have homework assignments which *will not* be collected. Homework will be assigned to practice the material and prepare for quizzes.

Examinations.

- (1) **Midterm Exam 1:** Friday, October 24, in the classroom,
- (2) **Midterm Exam 2:** Wednesday, November 19, in the classroom,
- (3) **Final Exam:** Friday, December 12 from 3pm-6pm, location TBA.

Grading Scheme 1.

	%
Homework	0%
Quizzes	12%
Exam 1	25%
Exam 2	25%
Final Exam	38%

Grading Scheme 2.

	%
Homework	0%
Quizzes	12%
max(Exam 1, Exam 2)	40%
Final Exam	48%

The grades posted on Course site are just the scores. They are *not necessarily* scaled to the values noted above. The grading scale is as follows:

Score Cutoffs	Course Grade
93	A
90	A–
87	B+
83	B
80	B–
77	C+
73	C
70	C–
67	D+
60	D
below 60	F

These thresholds may be adjusted at the end of the quarter, but only in a manner that will benefit students.

******Note: This syllabus is subject to change through the semester. Any updates to the syllabus will be announced in class.******

Course Coverage. The following is a list of lecture topics in the format of a tentative schedule. Each lecture corresponds to approximately 1-2 sections in the text.

Week	Sections
9/22	1.1 Sets, 3.1 Functions
9/29	2.2 Methods of Proof, 2.4 Mathematical induction, 3.2 Sequences and strings
10/06	3.3 Relations, 3.4 Equivalence relations, 3.5 matrices of relations, 6.1 Basic counting principles
10/13	6.2 Permutations and combinations, 6.3 Generalized permutations and combinations, 6.7 Binomial coefficients
10/20	6.8 Pigeonhole principle, Midterm 1, 7.1 Recurrence relations
10/27	7.2 Solving recurrence relations, 8.1 Examples of graphs, 8.2-8.3 Paths and cycles
11/03	8.4 Shortest-path algorithm, 8.5 Representation of graphs, 8.6 Isomorphisms of graphs
11/10	8.7 Planar graphs, 9.1 Examples of trees, 9.2 More trees
11/17	Midterm 2, 9.2 More trees
11/24	9.3-9.4 Minimal spanning trees, 9.5 Binary trees, Thanksgiving
12/01	9.8 Isomorphic trees
	Final on Dec 12

University Policies.

Academic Integrity. UCLA is a community of scholars. In this community, all members including faculty, staff and students alike are responsible for maintaining standards of academic honesty. As a student and member of the University community, you are here to get an education and are, therefore, expected to demonstrate integrity in your academic endeavors. You are evaluated on your own merits. Cheating, plagiarism, collaborative work, multiple submissions without the permission of the professor, or other kinds of academic dishonesty are considered unacceptable behavior and will result in formal disciplinary proceedings usually resulting in suspension or dismissal. See the Dean of Students website (<https://deanofstudents.ucla.edu/>) for more information. [source: Dean of Students syllabus statement (f033-476a-bccc-f6103528d959)]]

Accommodations for Students with Disabilities. If you are already registered with the Center for Accessible Education (CAE), please request your Letter of Accommodation in the Student Portal. If you are seeking registration with the CAE, please submit your request for accommodations via the CAE website. Students with disabilities requiring academic accommodations should submit their request for accommodations as soon as possible, as it may take up to two weeks to review the request. For more information, please visit the CAE website (<http://www.cae.ucla.edu/>) , visit the CAE at A255 Murphy Hall, or contact us by phone at (310) 825-1501. [source: Center for Accessible Education (Faculty Questions (<https://cae.ucla.edu/faculty/faqs>))]

Resources for Students. UCLA provides resources if you are feeling overwhelmed and need personal and/or academic assistance. Please see the Red Folder REV2020 web (<https://studentincrisis.ucla.edu/file/39679e1c-a57d-48d0-83a2-a906c1e53669>) for more information.

Title IX.

Advocacy and Confidential Services. Please note that Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking. If you have experienced sexual harassment or sexual violence, you can receive confidential support and advocacy at the CARE Advocacy Office for Sexual and Gender-Based Violence, 205 Covell Commons, Los Angeles, CA, 90095, care@careprogram.ucla.edu, (310) 206-246 5. Counseling and Psychological Services (CAPS) provides confidential counseling to all students and can be reached 24/7 at (310) 825-0768.

Reporting and Non-confidential Services. Your professor is required under the UC Policy on Sexual Violence and Sexual Harassment to inform the Title IX Coordinator should he become aware that you or any other student has experienced sexual violence or sexual harassment. In addition, You can also report sexual violence or sexual harassment directly to the University's Title IX Coordinator, 2255 Murphy Hall, titleix@equity.ucla.edu , (310) 206-3417. Reports to law enforcement can be made to UCPD at (310) 825-1491.